

WILSON WU

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Education

University of California, Los Angeles

Los Angeles, CA

Bachelor of Science in Electrical Engineering | GPA: 3.59 | Expected June 2029

- **Honors:** ECE Fast Track Honors Program (Top 20)
- **Relevant Coursework:** Introduction to Electrical Engineering, Internet of Things, Electricity and Magnetism, Object-Oriented Programming (C++), Data Structures and Algorithms

Technical Skills

Languages: Python, C/C++, Java

Tools/Technologies: KiCad, LTspice, PSpice, OSIRIS(Particle-in-Cell Simulation), Git, Docker, STM32, ESP32-S3

Scientific/Lab: NumPy, SciPy, Matplotlib, HDF5, Oscilloscope, DMM, Bench Power Supply, Soldering

Experience

UCLA Plasma Accelerator/Laser-Plasma Interactions Group

June 2026 – Present

Undergraduate Researcher | Python, OSIRIS(Particle-in-Cell Simulation), HDF5

Los Angeles, CA

- Investigated autoresonant beat-wave plasma excitation under Prof. Chan Joshi by implementing numerical differential-equation models of nonlinear plasma-wave amplitude growth and reproducing published results
- Configured particle-in-cell simulations in OSIRIS, tuning grid resolution and particle count to balance numerical accuracy and signal quality across multiple test scenarios
- Built a Python data pipeline to process 337 MB HDF5 simulation outputs, using FFT spectral analysis and Hilbert-transform envelope detection to extract wave dynamics

Bruin Supermileage (UCLA hydrogen fuel-cell vehicle team)

June 2026 – Present

Powertrain Project Engineer | KiCad, PSpice, LTspice, Power Electronics

Los Angeles, CA

- Designing a brake-chopper board to protect the DC bus from regenerative-braking voltage spikes by switching excess energy into an off-board braking resistor before the 60 V competition limit
- Defined system requirements from powertrain architecture and competition rules, specifying a hysteresis trigger band, 70 V component rating, 3.2 kW dissipation target, and switching response under 2 microseconds

Low Voltage Engineer | KiCad, LTspice, CAN, Embedded Systems

October 2025 – June 2026

- Designed a two-layer KiCad PCB for the vehicle safety system, integrating hydrogen-leak detection, ignition interlock, driver emergency-stop logic, and CAN bus communication to meet Shell Eco-marathon safety requirements
- Engineered a single-fault-tolerant hardware interlock combining hydrogen-sensor, ignition-signal, and emergency-stop inputs to guarantee power isolation independently of semiconductor fault modes
- Verified the safety board through bench and vehicle-level integration testing, confirming CAN bus connectivity and validating relay actuation across all emergency-stop and sensor-fault conditions

Projects

Autonomous Line-Following Car | C++, PID Control, Sensors

Spring 2026

- Developed and tuned proportional-derivative control for an autonomous line-following car, iterating on controller gains across multiple vehicle speeds to reduce oscillation and improve track stability
- Diagnosed sensor-alignment bias through serial-data testing and implemented calibration offsets and turn-detection logic to improve autonomous track following

Gacha Thrift | ESP32-S3, C++, Python

February 2026

- Built a gacha-style thrift-clothing dispenser in 36 hours, integrating mechanical, electrical, and software subsystems into a functional end-to-end product
- Implemented closed-loop ESP32-S3 motor control using encoder feedback and synchronized physical dispensing with a browser-based animation through a Python WebSocket bridge

Micromouse: Autonomous Maze-Solving Robot | KiCad, STM32, C

December 2025 – Present

- Led a team developing an autonomous maze-solving robot, coordinating hardware and firmware work across power, sensing, and motor-control subsystems
- Designed a mixed-signal PCB integrating four infrared emitter-detector pairs and a gyroscope, routing low-amplitude sensor traces away from high-current motor-driver traces to protect signal integrity
- Diagnosed post-fabrication layout errors and restored full functionality through jumper-wire patches and STM32 firmware corrections
- Implemented a flood-fill maze-solving algorithm in C on an STM32 microcontroller, mapping traversed cells and dynamically updating costs to converge on the optimal path